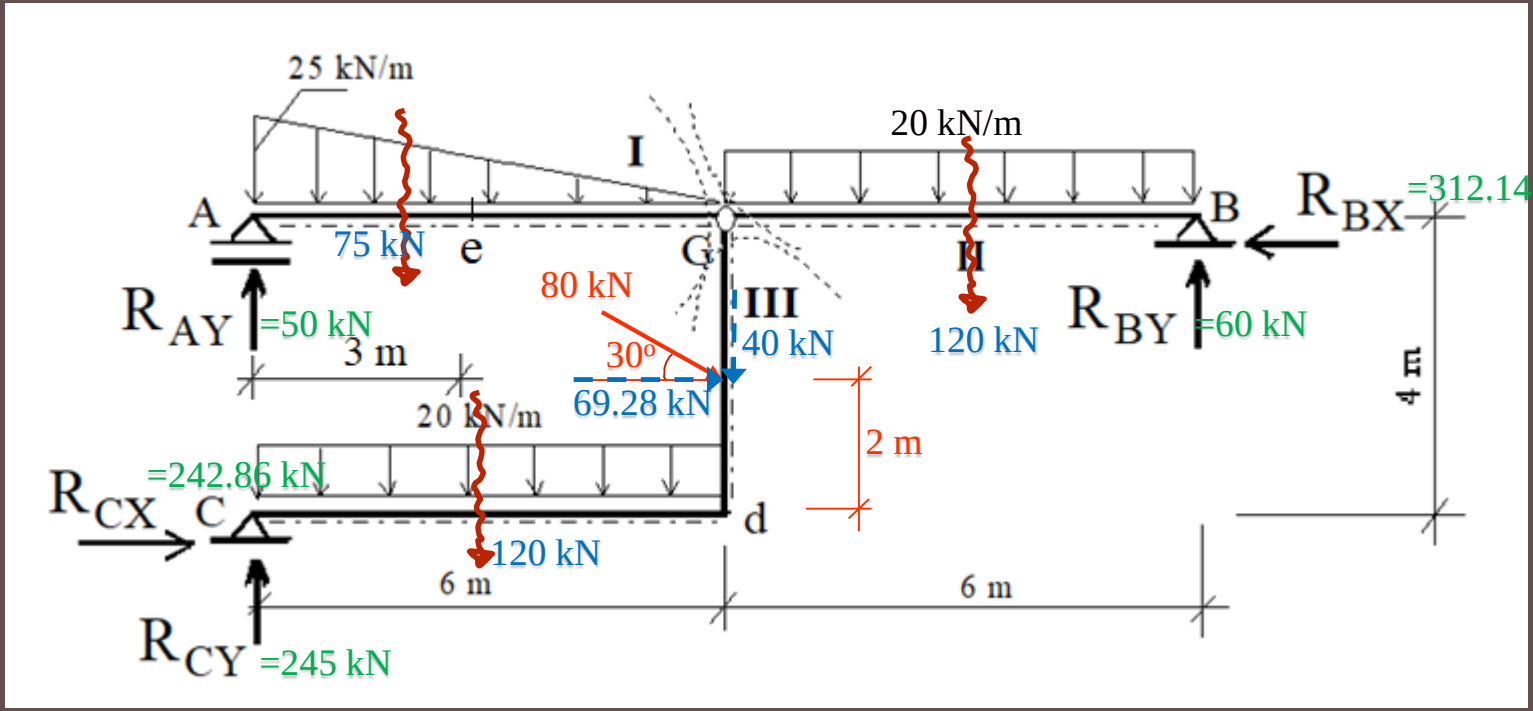
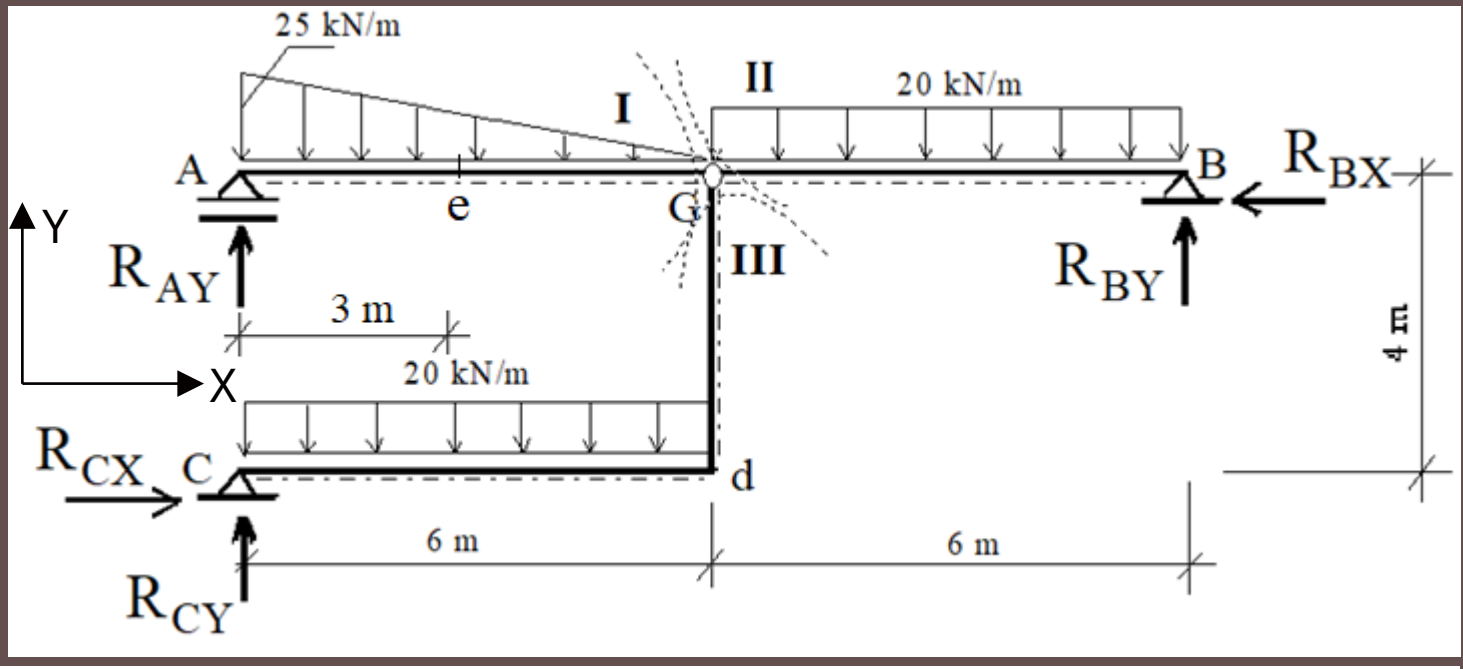
Kısa sınav.1



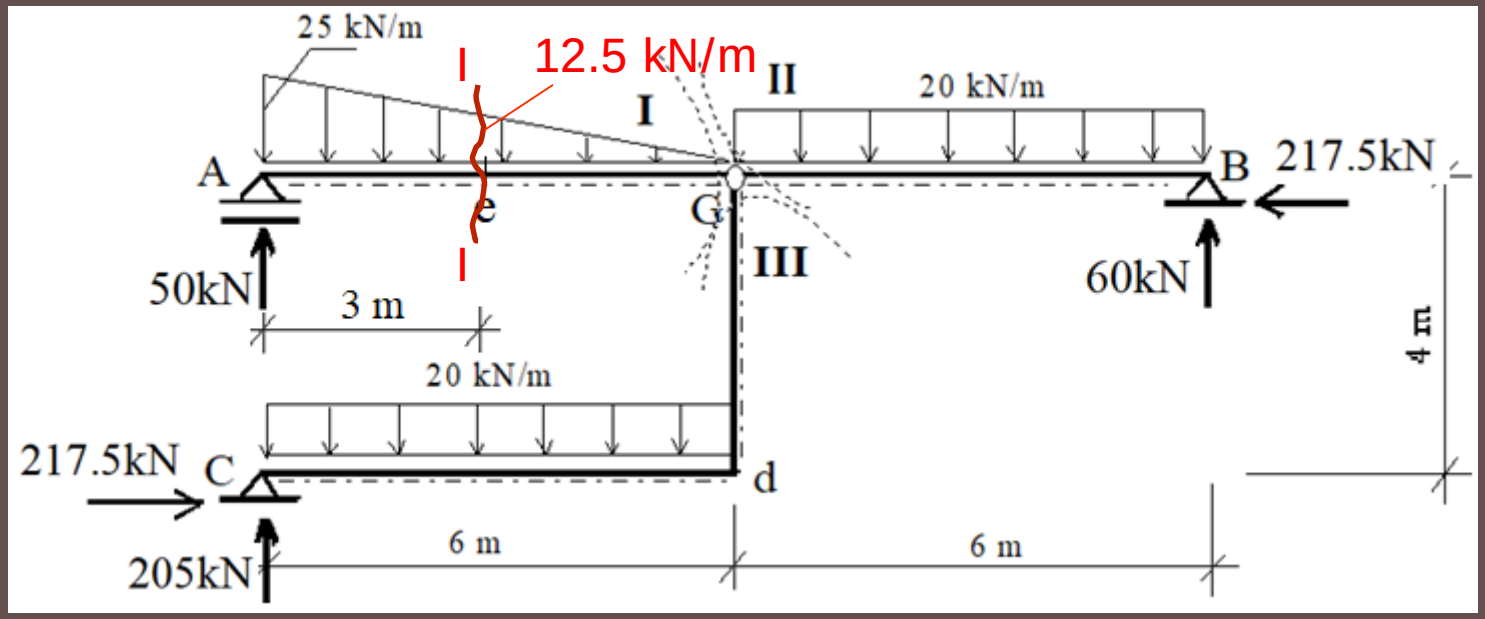
Soru: Mesnet tepkilerini hesaplayınız.

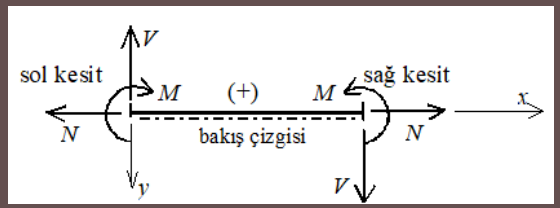
«e, d ve G» kesitlerindeki kesit tesirlerinin hesabı



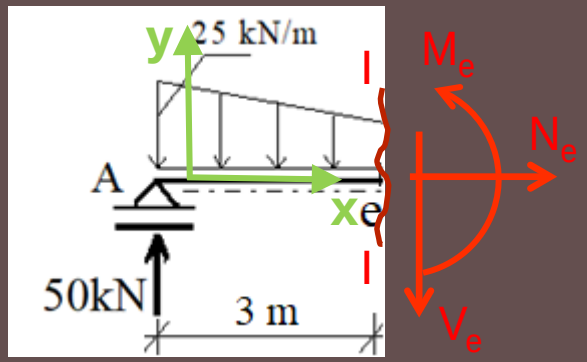
Tüm sistem	$\sum X = 0$	$\rightarrow$	$R_{BX} = R_{CX}$	
(I + II + III)	$\sum Y = 0$	$\rightarrow$	$R_{CY} + R_{BY} + R_{AY} - 20 \cdot 12 - 25 \cdot 3 = 0$	(1)
(I)	$\sum M_G = 0$	$\rightarrow$	$R_{AY} \cdot 6 - 25 \cdot 3 \cdot 4 = 0 \Rightarrow R_{AY} = 50\text{kN}$	
(II)	$\sum M_G = 0$	$\rightarrow$	$R_{BY} \cdot 6 - 20 \cdot 6 \cdot 3 = 0 \Rightarrow R_{BY} = 60\text{kN}$	
(III)	$\sum M_G = 0$	$\rightarrow$	$-R_{CX} \cdot 4 + R_{CY} \cdot 6 - 20 \cdot 6 \cdot 3 = 0$	(2)

R_{AY}, R_{BY} nin değerleri (1) de yerine konulduğunda $R_{CY} = 205\text{kN}$ *olarak, bu değer (2) de yerine konulduğunda ise* $R_{CX} = 217.5\text{kN}$ *olarak bulunmuş olur.*





«e» kesitindeki kesit tesirleri (I-I kesimi)

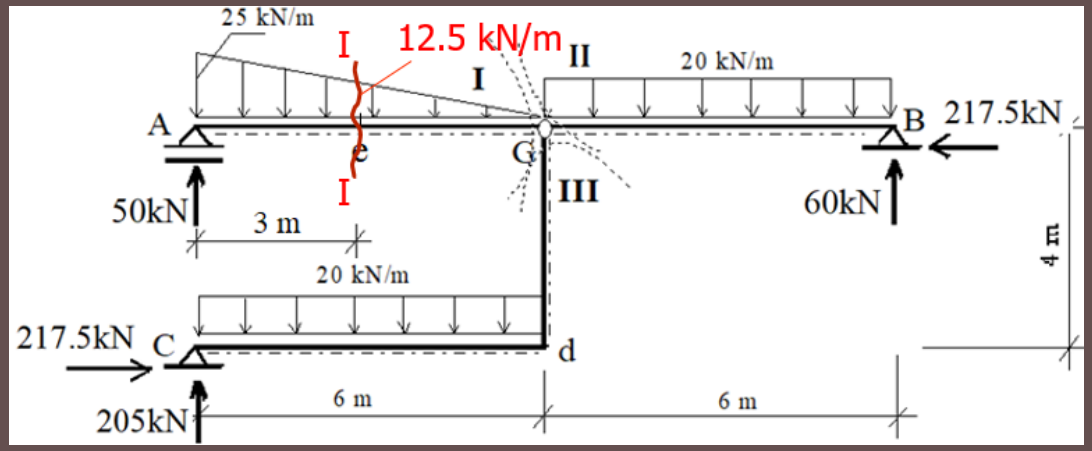


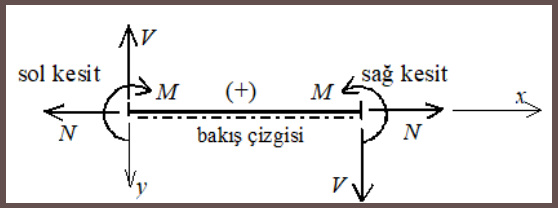
$$\sum x = 0;\ N_e = 0$$

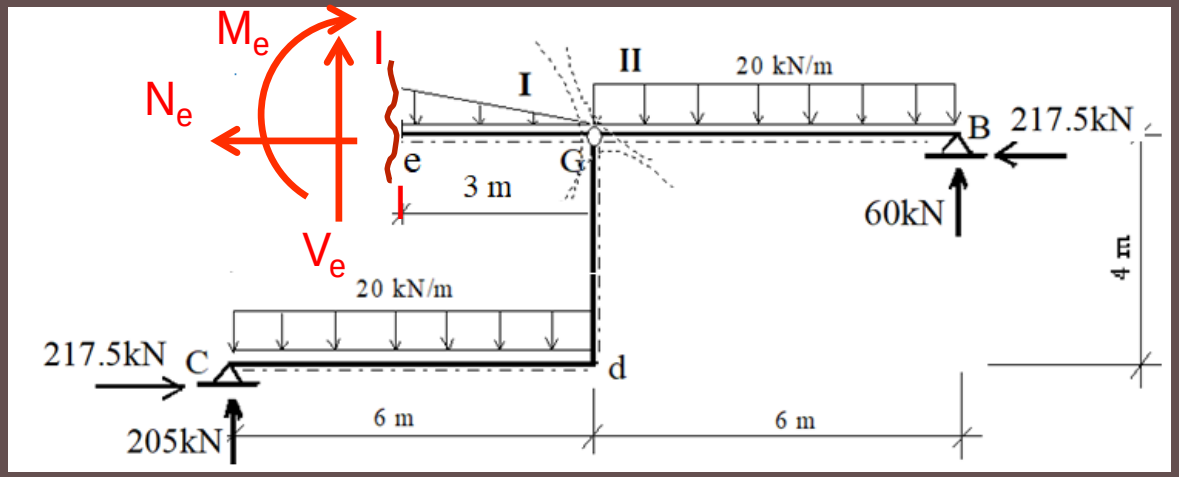
$$\sum y = 0;\ V_e = 50 - 12.5\cdot 3 - 12.5\cdot 3\cdot 0.5 = -6.25\ \text{kN}$$

$$\sum M_e = 0;\ M_e = 50\cdot 3 - 12.5\cdot 3\cdot 1.5 - 12.5\cdot 3\cdot 0.5\cdot 2 = 56.25\ \text{kNm}$$

VEYA



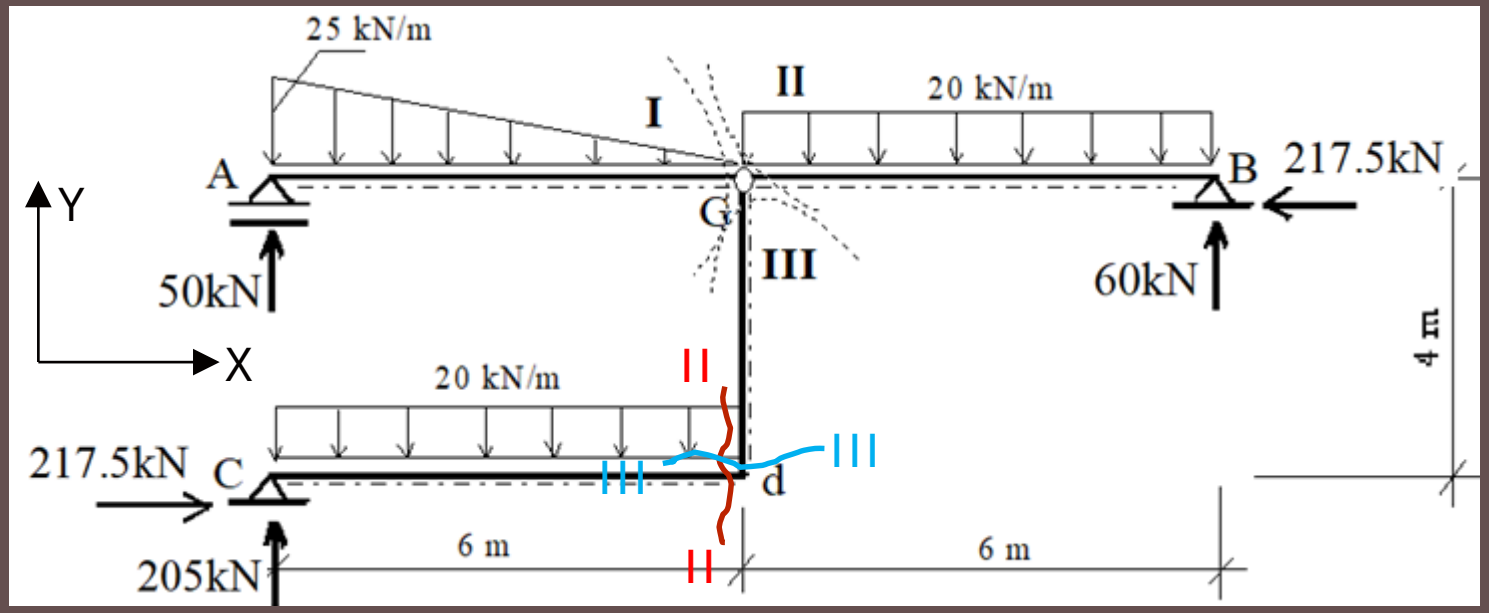


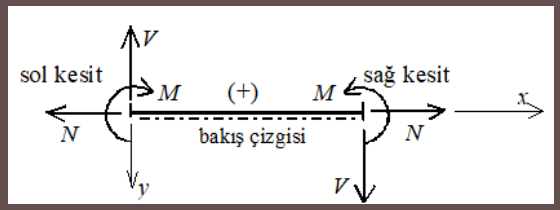


$$\sum x = 0;\ N_e = 217.5 - 217.5 = 0$$

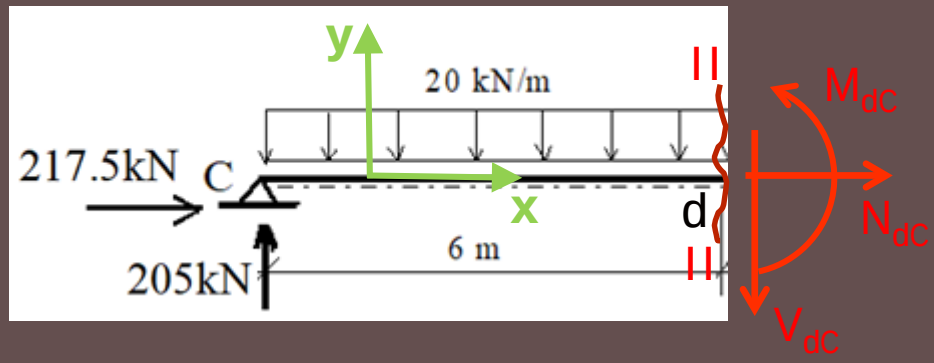
$$\sum y = 0;\ V_e = -205 + 20\cdot 6 + 12.5\cdot 3\cdot 0.5 + 20\cdot 6 - 60 = -6.25\ \text{kN}$$

$$\sum M_e = 0;\ M_e = -205\cdot 3 + 217.5\cdot 4 - 12.5\cdot 3\cdot 0.5\cdot 1 - 20\cdot 6\cdot 6 + 60\cdot 9 = 56.25\text{kNm}$$





«d» kesiti soluna ait kesit tesirleri (II-II kesimi)

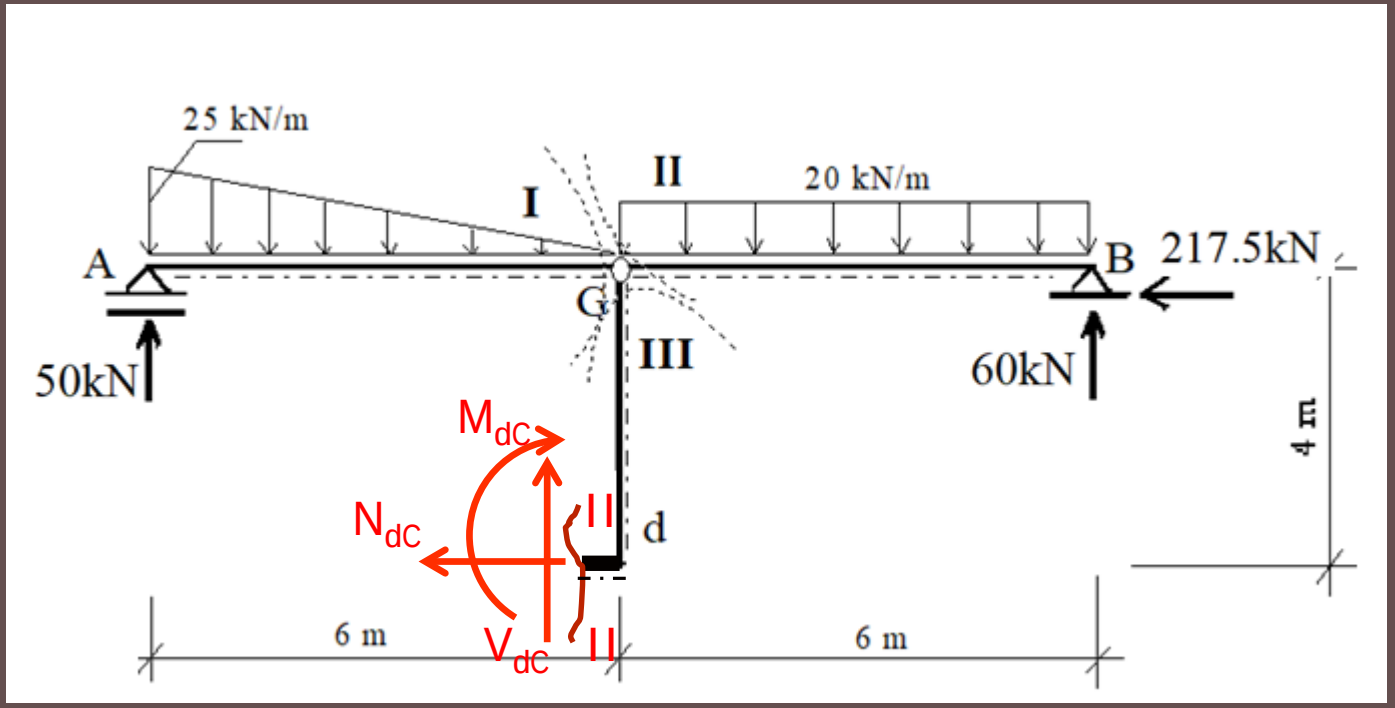


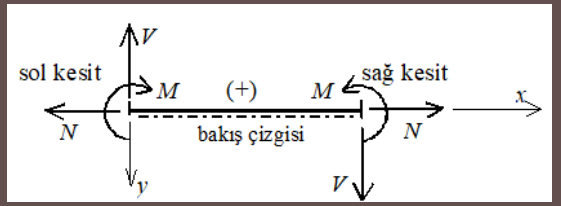
$$\sum x = 0;\ N_{dC} = -217.5\,kN$$

$$\sum y = 0;\ V_{dC} = 205 - 20 \cdot 6 = 85\,kN$$

$$\sum M_d = 0;\ M_{dC} = 205 \cdot 6 - 20 \cdot 6 \cdot 3 = 870\,kNm$$

VEYA

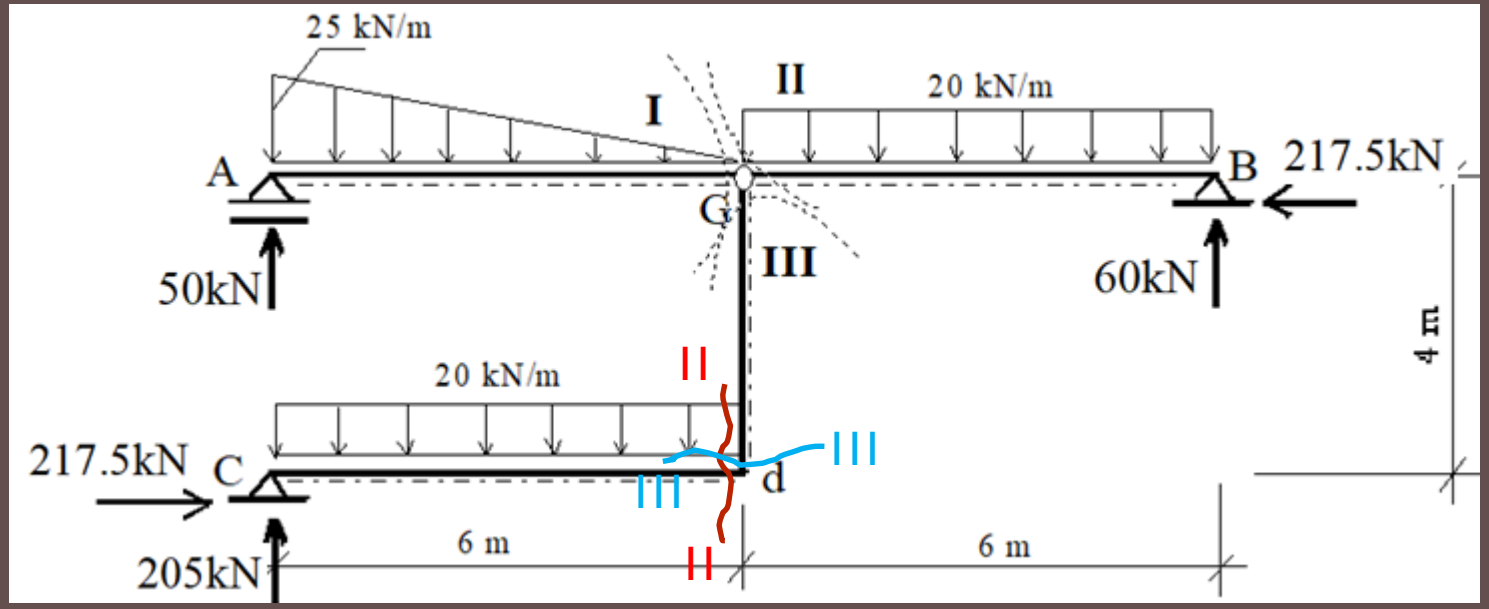




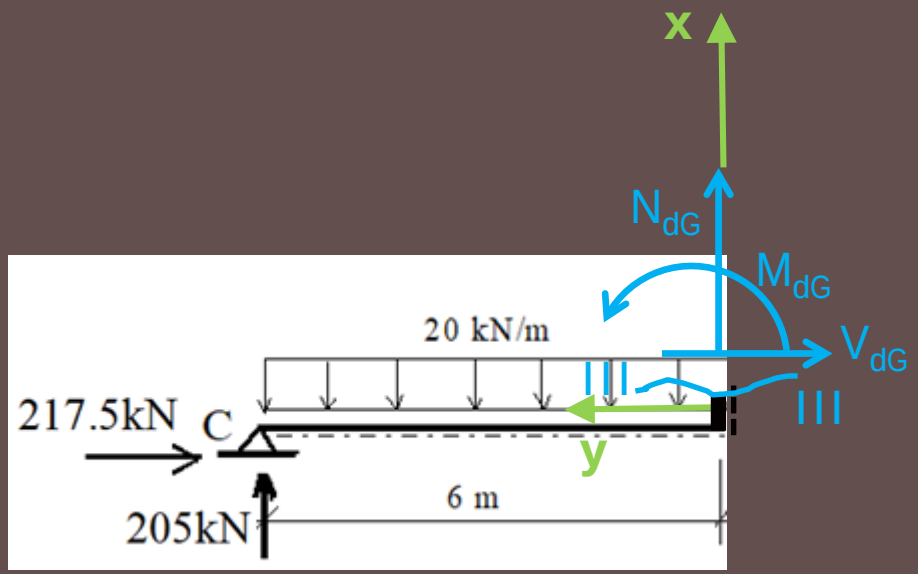
$$\sum x = 0;\ N_{dC} = -217.5\,kN$$

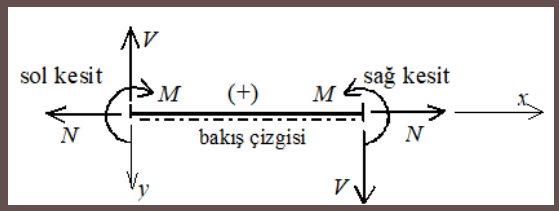
$$\sum y = 0;\ V_{dC} = -50 + 25 \cdot 6 \cdot 0.5 + 20 \cdot 6 - 60 = 85\,kN$$

$$\sum M_d = 0;\ M_{dC} = -50 \cdot 6 + 25 \cdot 6 \cdot 0.5 \cdot 4 - 20 \cdot 6 \cdot 3 + 60 \cdot 6 + 217.5 \cdot 4 = 870\,kNm$$



«d» kesiti sağına ait kesit tesirleri (III-III kesimi)



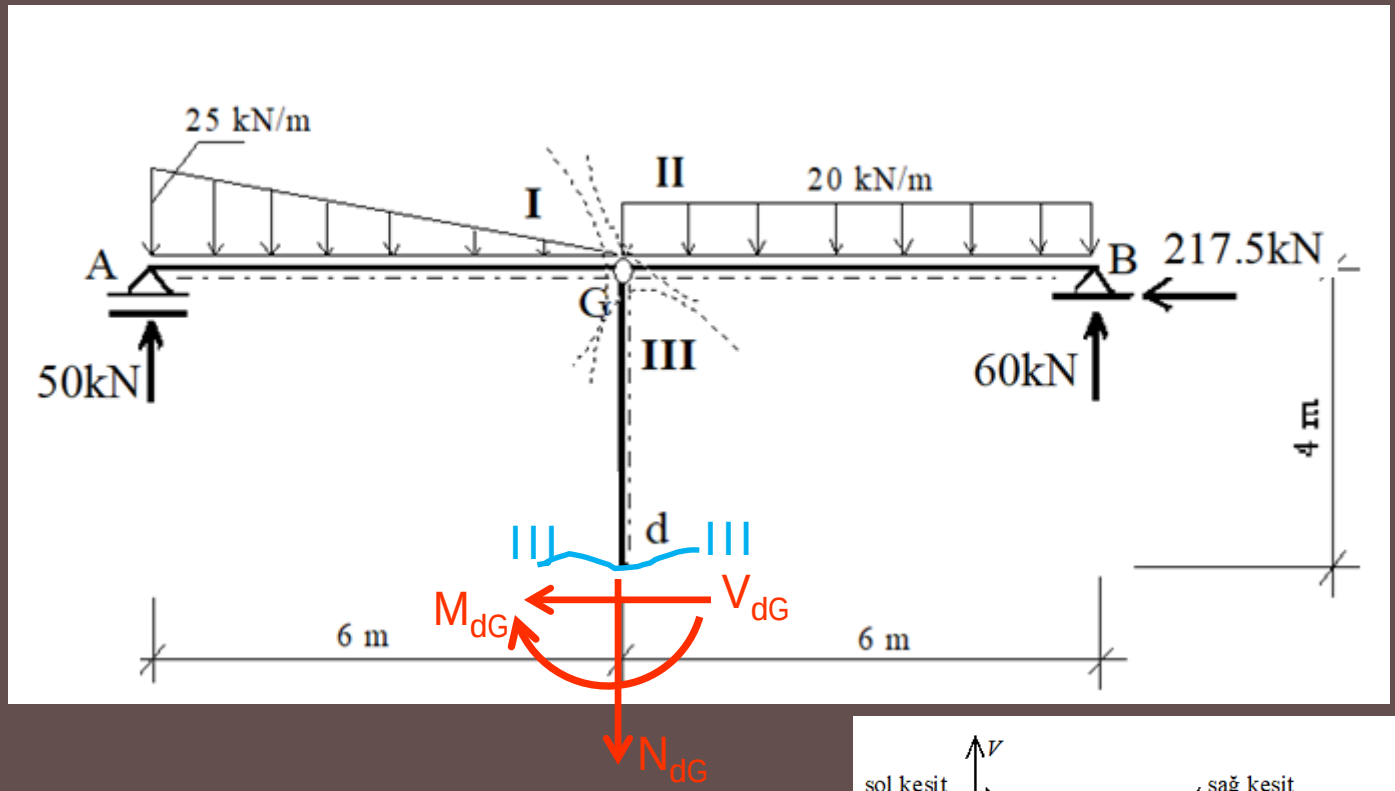


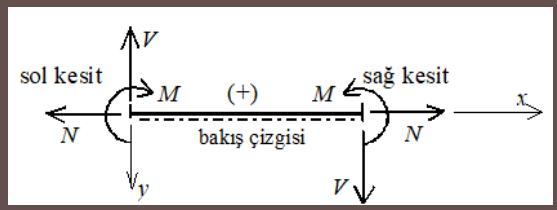
$$\sum x = 0;\ N_{dG} = -205 + 20 \cdot 6 = -85\,kN$$

$$\sum y = 0;\ V_{dG} = -217.5\,kN$$

$$\sum M_d = 0;\ M_{dG} = 205 \cdot 6 - 20 \cdot 6 \cdot 3 = 870\,kNm$$

VEYA





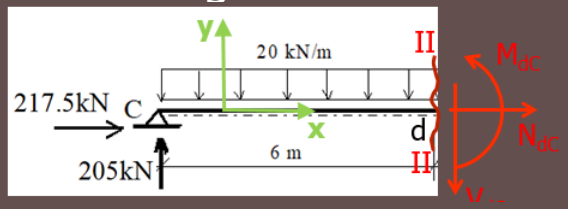
$$\sum x = 0;\ N_{dG} = 50 - 25 \cdot 6 \cdot 0.5 - 20 \cdot 6 + 60 = -85\,kN$$

$$\sum y = 0;\ V_{dG} = -217.5\,kN$$

$$\sum M_d = 0;\ M_{dG} = 217.5 \cdot 4 + 60 \cdot 6 - 20 \cdot 6 \cdot 3 + 25 \cdot 6 \cdot 0.5 \cdot 4 - 50 \cdot 6 = 870\,kNm$$

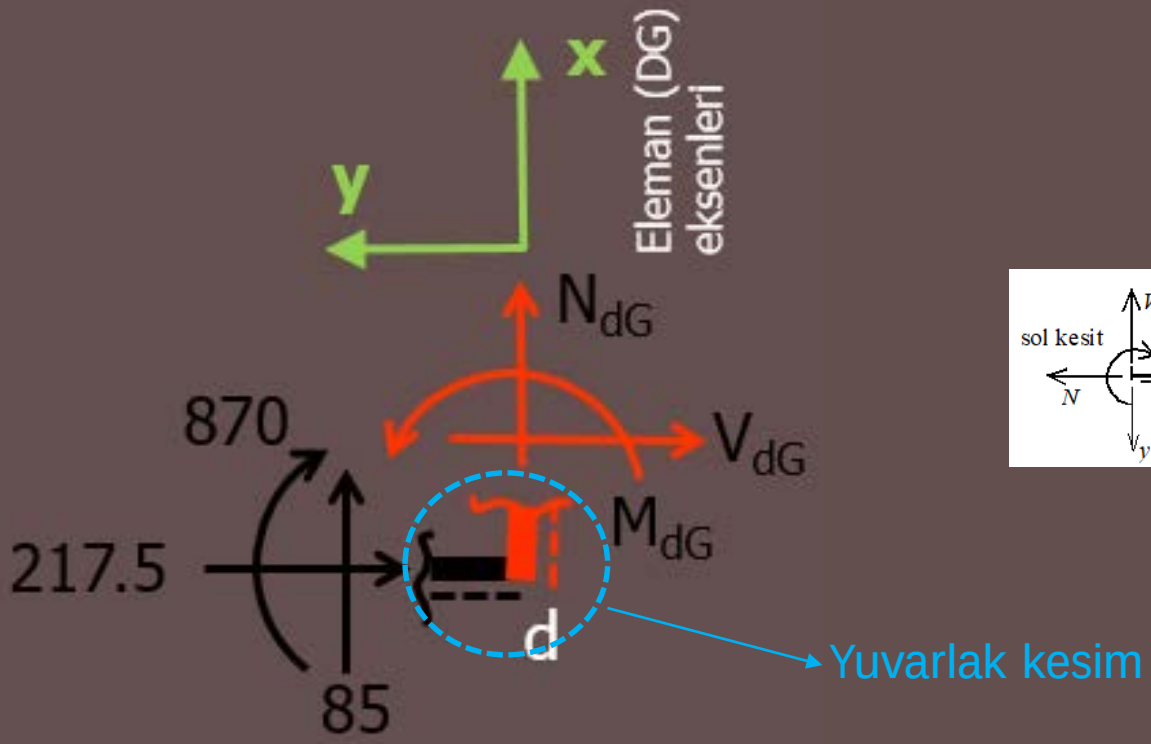
Bir çubuğun/kesitin sağ veya sol tarafına ait bilinen kesit tesirinden yararlanarak diğer tarafına ait kesit tesirlerinin hesabı

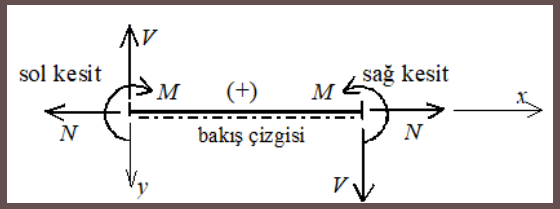
«d» kesitinin soluna ait kesit tesirleri hesaplanmış olsun; N_{dC}= -217.5 kN, V_{dC}= 85 kN, M_{dC}= 870 kNm. Kesitin sağ tarafına ait kesit tesirlerini hesaplayabilmek için;



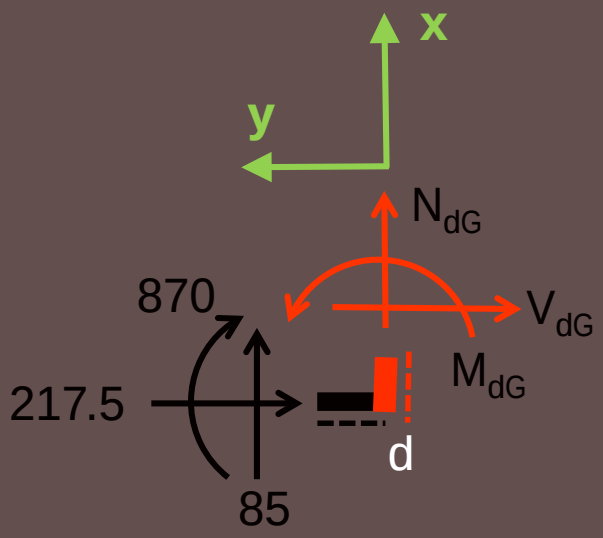
$$\sum x = 0;\ N_{dC} = -217.5\,kN$$
$$\sum y = 0;\ V_{dC} = 205 - 20\cdot 6 = 85\,kN$$
$$\sum M_d = 0;\ M_{dC} = 205\cdot 6 - 20\cdot 6\cdot 3 = 870\,kNm$$

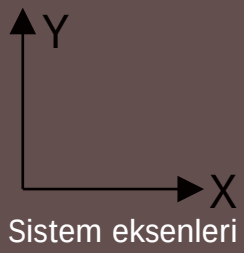
1. «d» kesiti civarında ve çok yakınında yuvarlak bir kesim yapılır.
2. Kesimle açığa çıkan ve değerleri bilinen kesit tesirleri bir dış yük gibi kesite etki ettirilir.





3. Sistem eksenlerine göre denge şartları yazılarak kesitin diğer tarafına ait bilinmeyen kesit tesirleri hesaplanır.



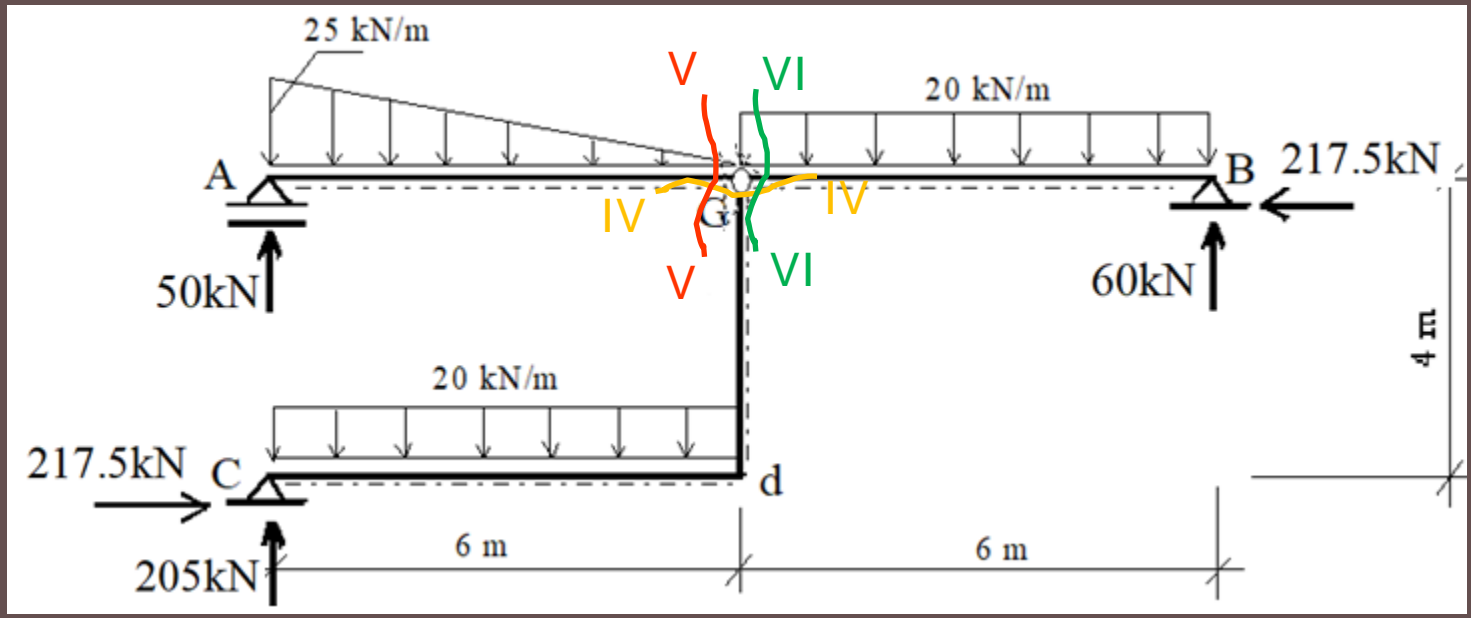


$\Sigma X=0;\ V_{dG} = -217.5$ kN

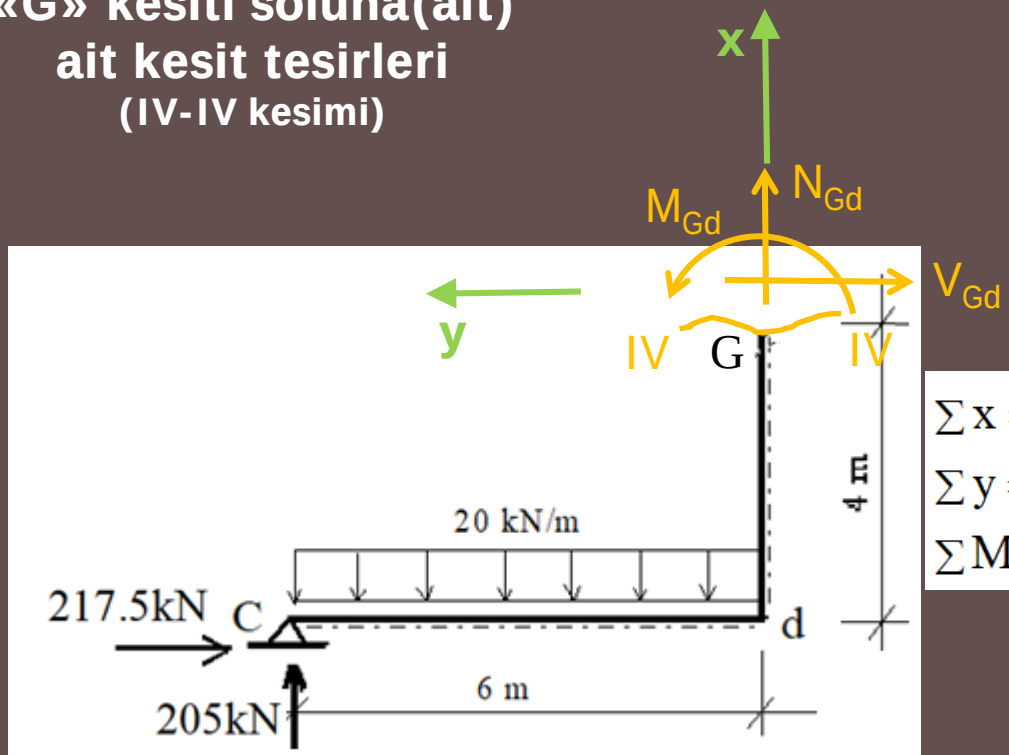
$\Sigma Y=0;\ N_{dG} = -85$ kN

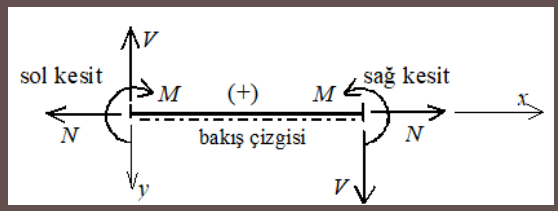
$\Sigma M_d=0;\ M_{dG} = 870$ kNm

«G» kesitine ait kesit tesirleri



«G» kesiti soluna(alt) ait kesit tesirleri
(IV-IV kesimi)



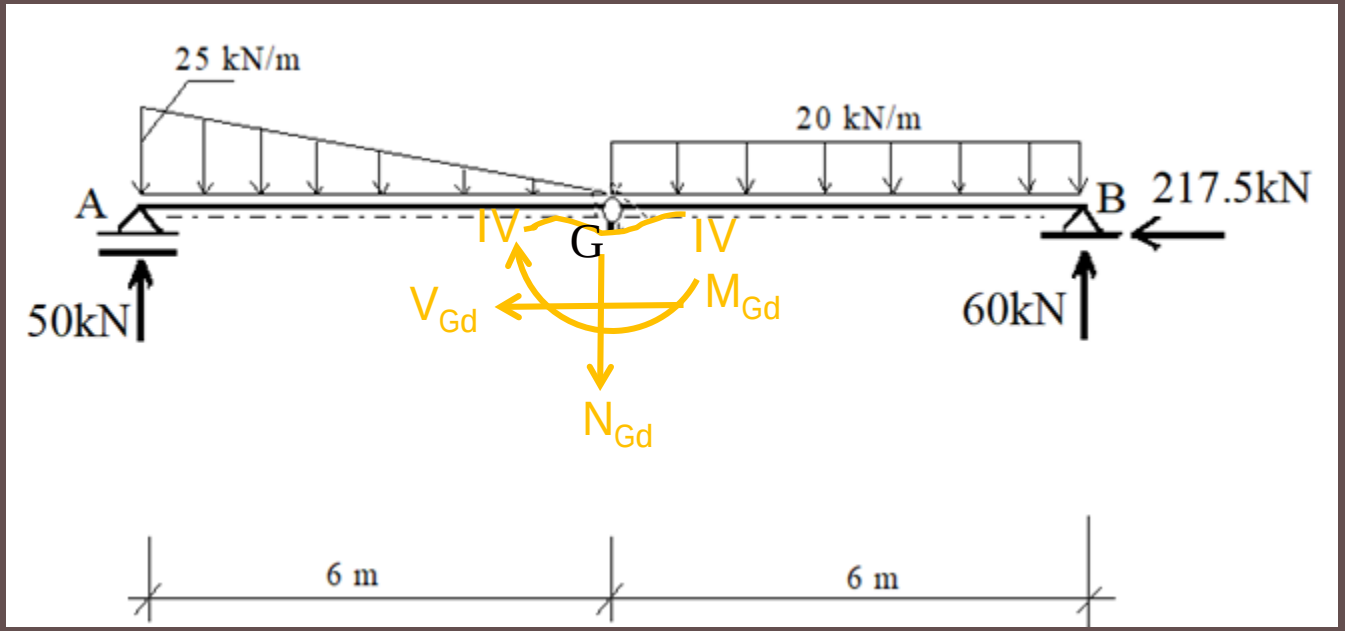


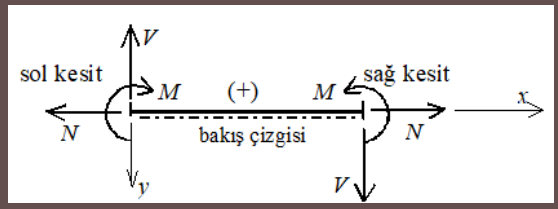
$$\sum x = 0;\ N_{Gd} = -205 + 20 \cdot 6 = -85\text{kN}$$

$$\sum y = 0;\ V_{Gd} = -217.5\text{kN}$$

$$\sum M_G = 0;\ M_{Gd} = -217.5 \cdot 4 - 20 \cdot 6 \cdot 3 + 205 \cdot 6 = 0$$

VEYA



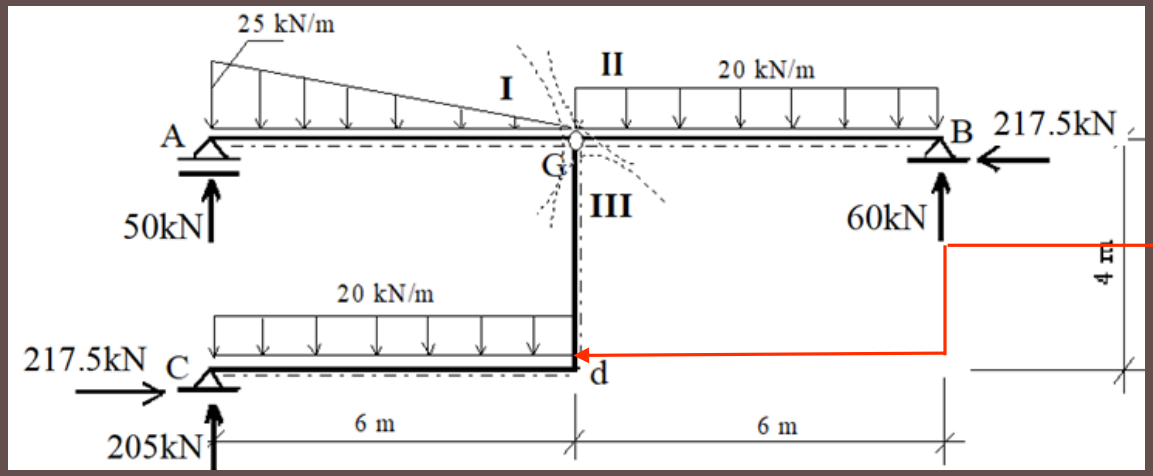


$$\sum x = 0;\ N_{Gd} = 50 - 25\cdot 6\cdot 0.5 - 20\cdot 6 + 60 = -85\,kN$$

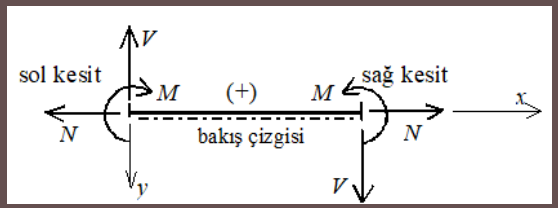
$$\sum y = 0;\ V_{Gd} = -217.5\,kN$$

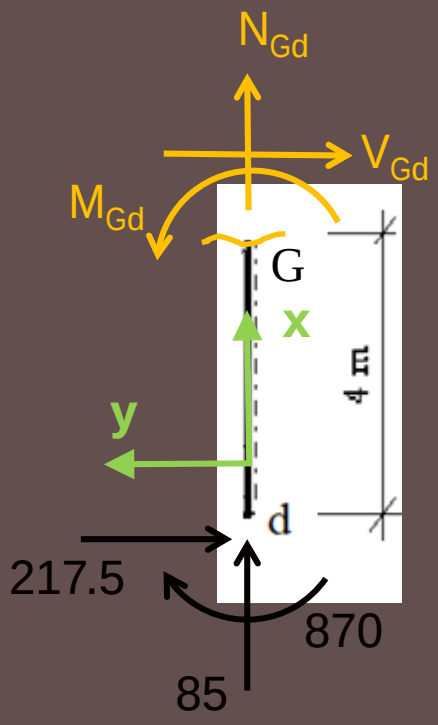
$$\sum M_G = 0;\ M_{Gd} = -50\cdot 6 + 25\cdot 6\cdot 0.5\cdot 4 - 20\cdot 6\cdot 3 + 60\cdot 6 = 0$$

VEYA

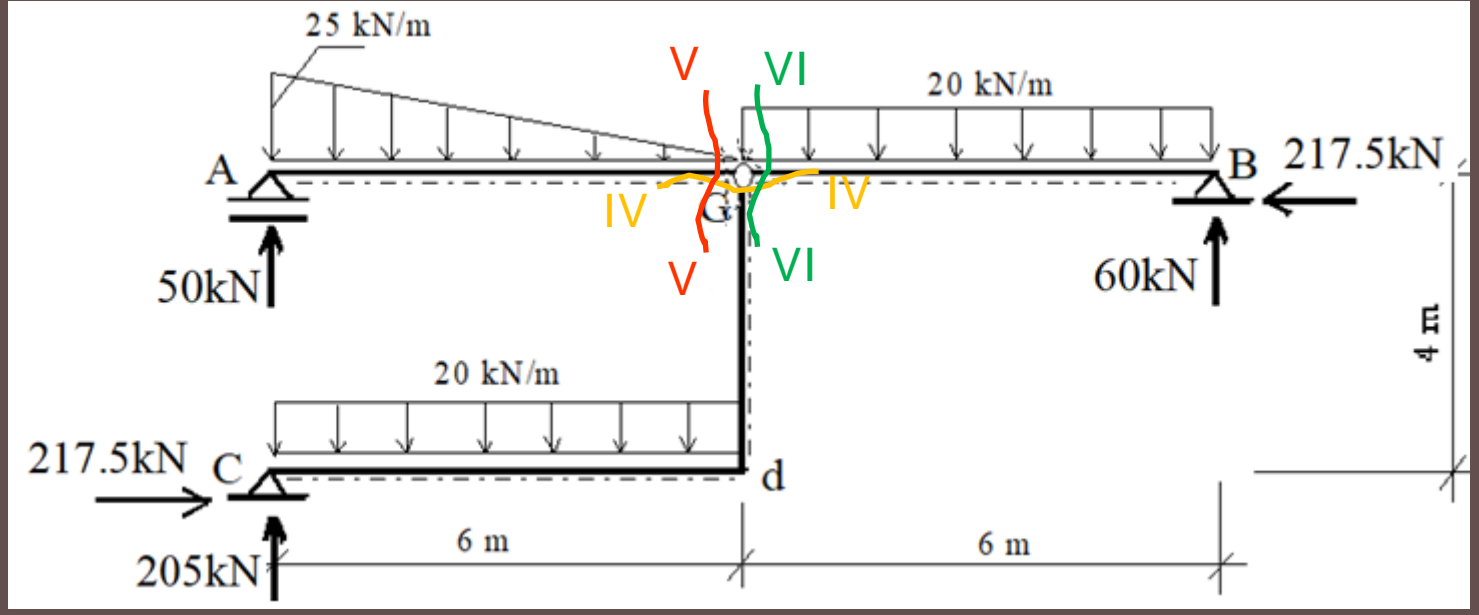


$$N_{dG} = -85\,kN$$
$$V_{dG} = -217.5\,kN$$
$$M_{dG} = 870\,kNm$$



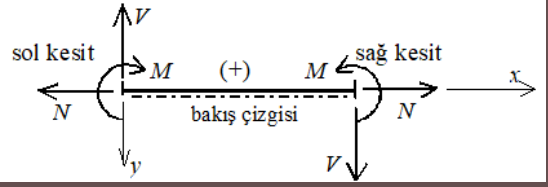


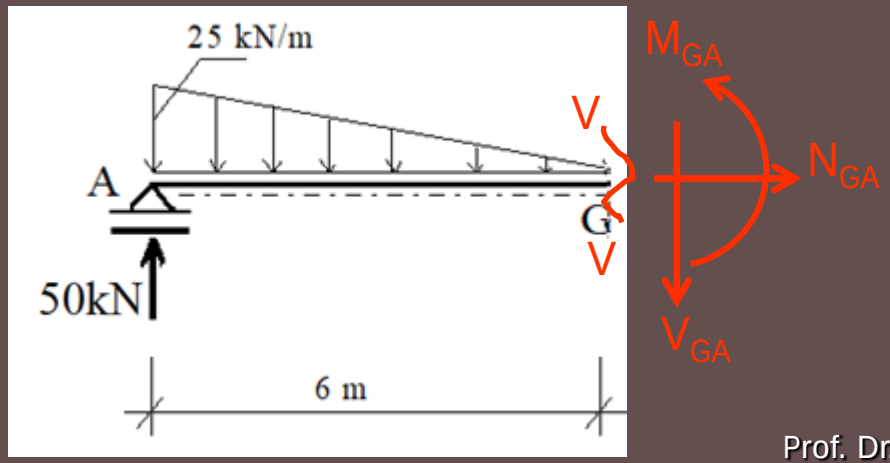
$$\sum x = 0;\ N_{Gd} = -85kN$$
$$\sum y = 0;\ V_{Gd} = -217.5kN$$
$$\sum M_G = 0;\ M_{Gd} = 870 - 217.5 \cdot 4 = 0$$



«G» kesiti soluna(üst) ait kesit tesirleri

(V-V kesimi)

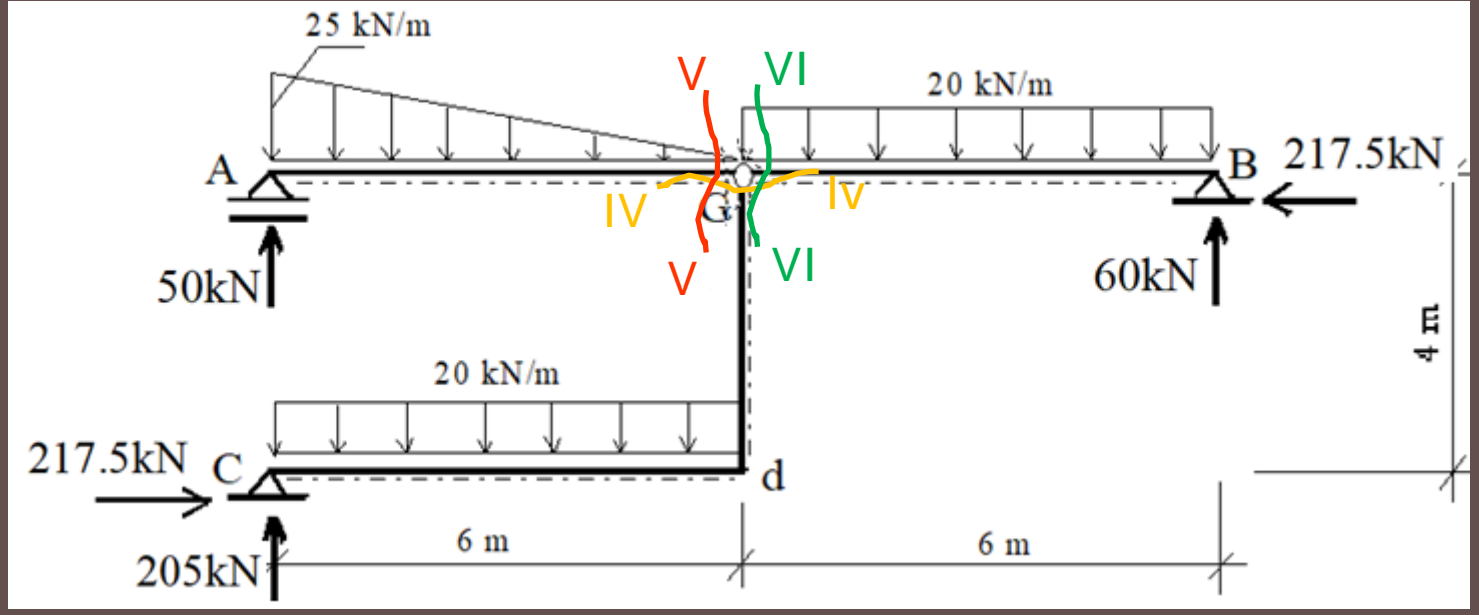




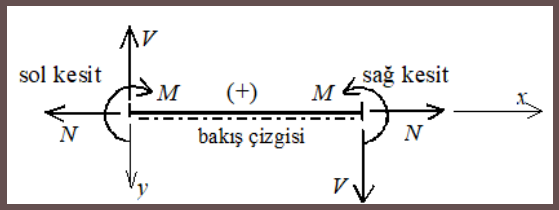
$$\sum x = 0;\ N_{GA} = 0$$

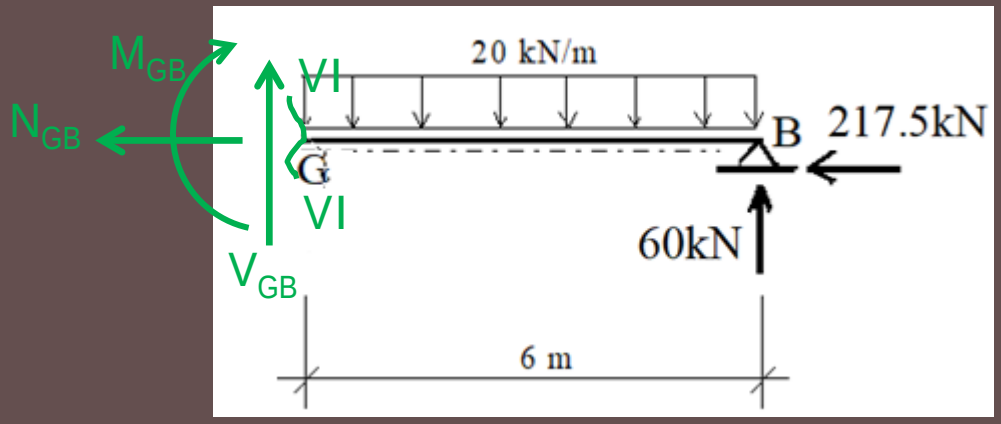
$$\sum y = 0;\ V_{GA} = 50 - 25 \cdot 6 \cdot 0.5 = -25\,\text{kN}$$

$$\sum M_G = 0;\ M_{GA} = 50 \cdot 6 - 25 \cdot 6 \cdot 0.5 \cdot 4 = 0$$



«G» kesiti sağına(üst) ait kesit tesirleri
(VI-VI kesimi)



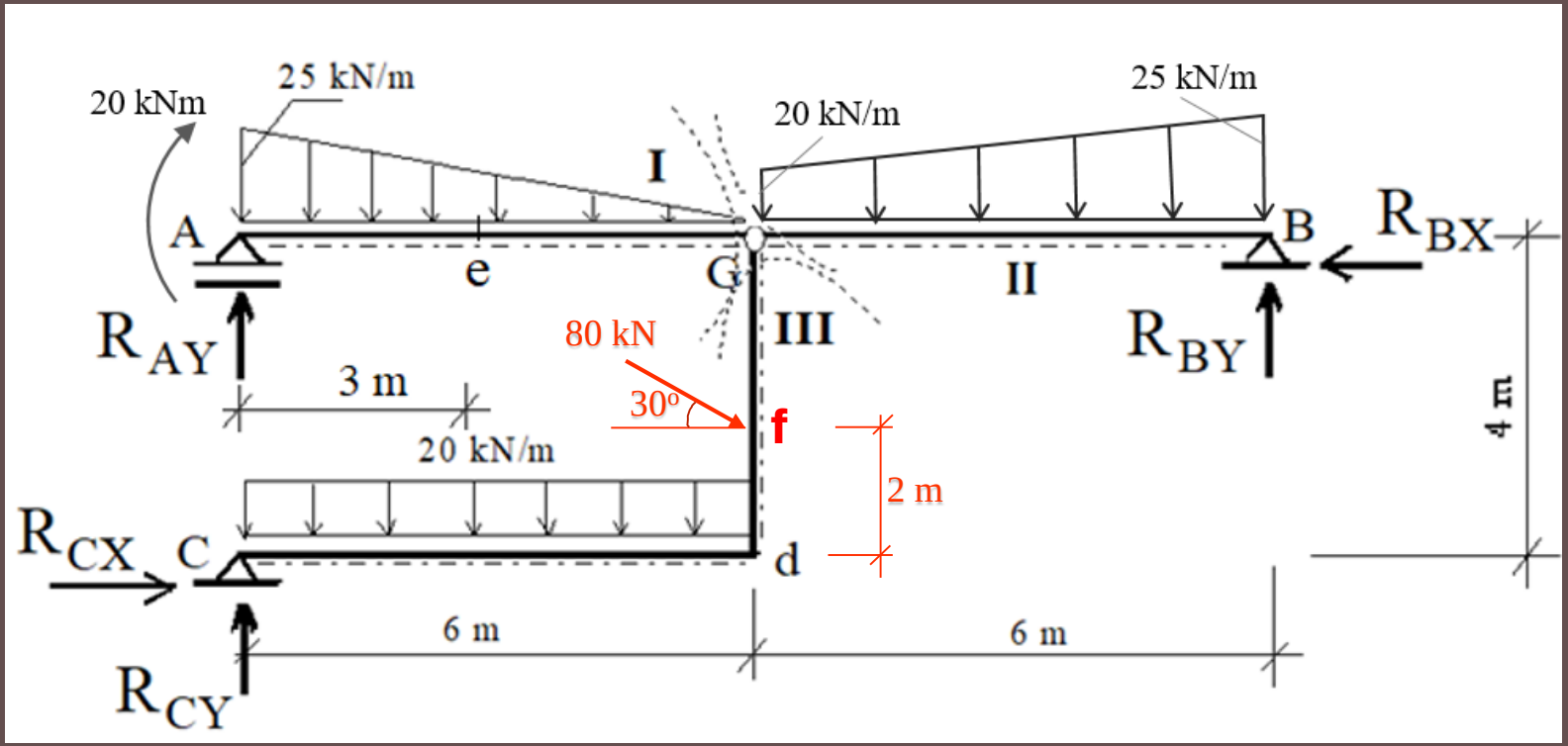


$$\sum x = 0;\ N_{GB} = -217.5\,\text{kN}$$

$$\sum y = 0;\ V_{GB} = 20 \cdot 6 - 60 = 60\,\text{kN}$$

$$\sum M_G = 0;\ M_{GB} = 60 \cdot 6 - 20 \cdot 6 \cdot 3 = 0$$

Kısa Sınav.2

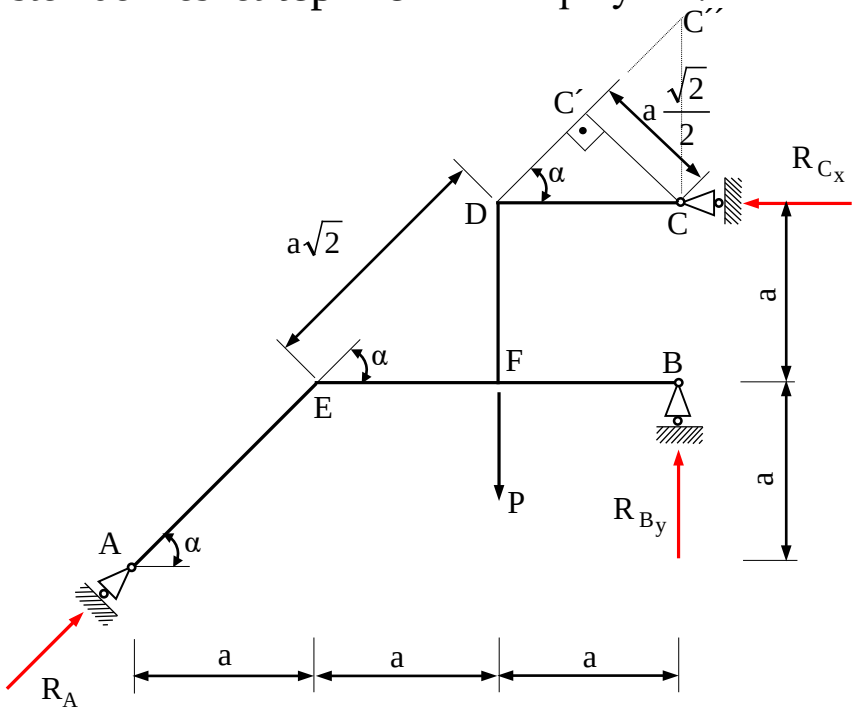


Soru:

1. Mesnet tepkilerini hesaplayınız.
2. «f» kesitindeki kesit tesirlerini hesaplayınız.

Örnek 3.7

Şekil 3.24'de yükleme durumu ve boyutları verilen sistemde mesnet tepkilerini hesaplayınız.



$$2CC''^2 = a^2$$

$$CC' = a\frac{\sqrt{2}}{2}$$

$$\cos\alpha = \sqrt{2}/2$$

$$\sin\alpha = \sqrt{2}/2$$

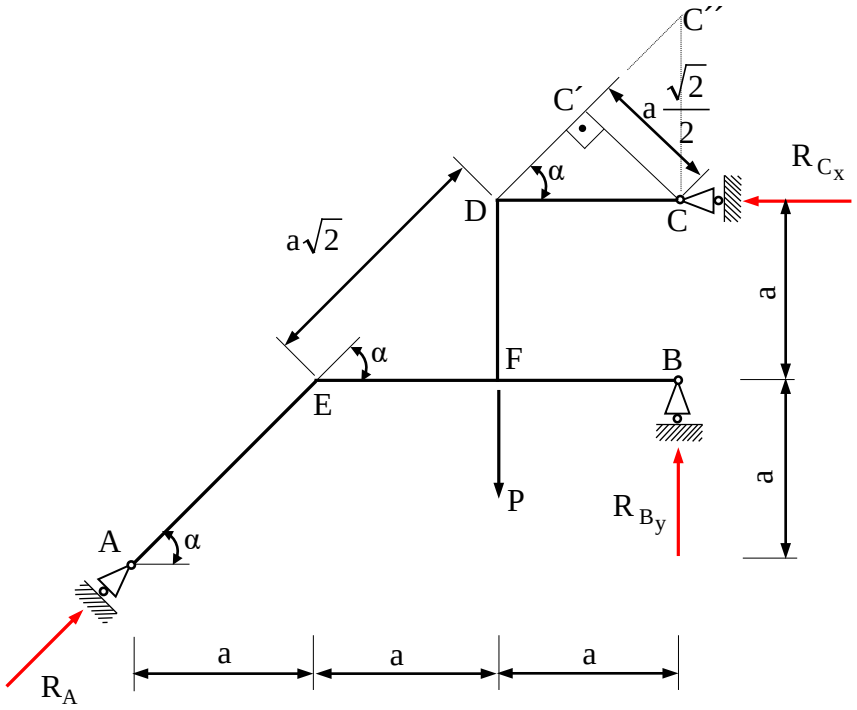
Şekil 3.24

Burada önce R_A ve R_{C_x} 'in kesiştiği D noktasına göre

$$\sum M_D = 0 \longrightarrow R_{B_y} = 0$$

R_{B_y} ve R_{C_x}'in kesiştiği C noktasına göre

$$\sum M_C = 0 \longrightarrow R_A \cdot a\frac{\sqrt{2}}{2} - P\cdot a = 0 \longrightarrow R_A = \sqrt{2}.P$$



$2CC''^2 = a^2$

$CC' = a\frac{\sqrt{2}}{2}$

$\cos\alpha = \sqrt{2}/2$

$\sin\alpha = \sqrt{2}/2$

$R_A = \sqrt{2}.P$

$R_{By} = 0$

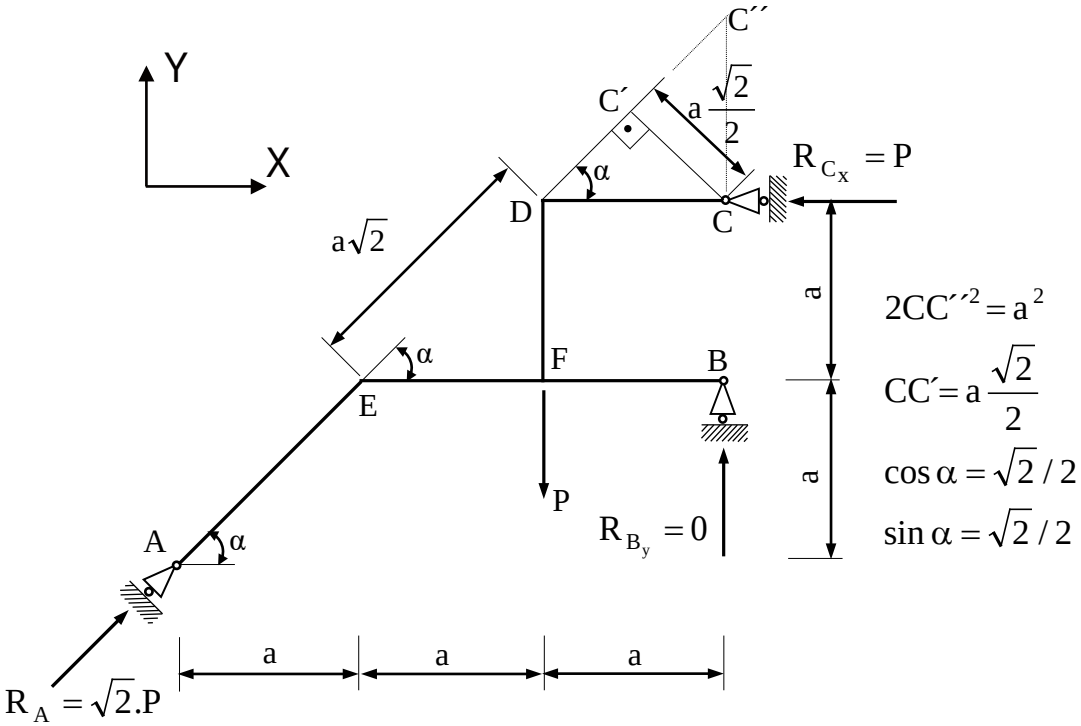
Şekil 3.24

R_A ile R_{By}'nin kesiştiği C´´ noktasına göre

$$\sum M_{C''} = 0 \longrightarrow R_{Cx}\,a - P\cdot a = 0 \longrightarrow R_{Cx} = P$$

Görüldüğü gibi birer bilinmeyenli 3 denge şartından üç bilinmeyen kolay bir şekilde hesaplanmıştır.

Kontrol:



Şekil 3.24

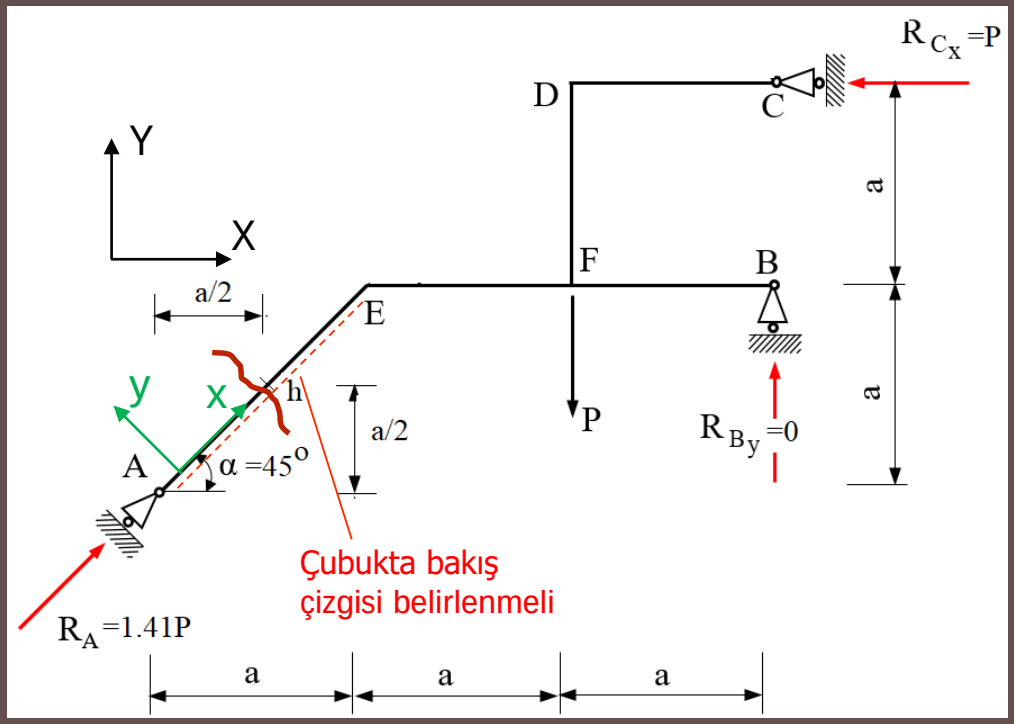
$$\sum P_x = \sum X = 0 \longrightarrow R_A \cos\alpha - R_{C_x} = 0 \longrightarrow \sqrt{2} \cdot P \cdot \frac{\sqrt{2}}{2} - P = 0$$

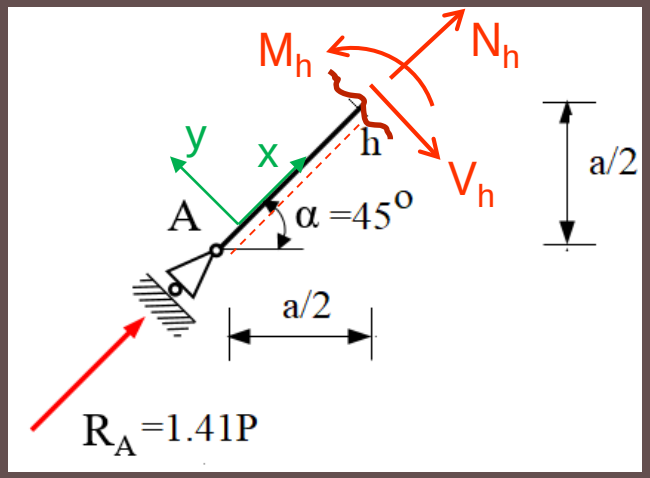
$$\sum P_y = \sum Y = 0 \longrightarrow R_A \sin\alpha - P - R_{B_y} = 0 \longrightarrow \sqrt{2}P \cdot \frac{\sqrt{2}}{2} - P - 0 = 0$$

Eğer $\sum X = 0$, $\sum Y = 0$ ve $\sum M_A = 0$ denge şartları yazılsaydı üç adet iki bilinmeyenli denklemi çözmek gerekirdi. Buradaki seçimde ise mesnet tepkilerinin her biri, hem bir bilinmeyenli birer eşitlikten ve hem de hesaplanan bilinmeyen kullanılmadan hesaplanmıştır.

«h» Kesitindeki kesit tesirlerinin hesabı

I. Seçenek: Kesimin solunda kalan sistem parçası üzerinde eleman eksenlerine göre denge şartları yazılacaktır;



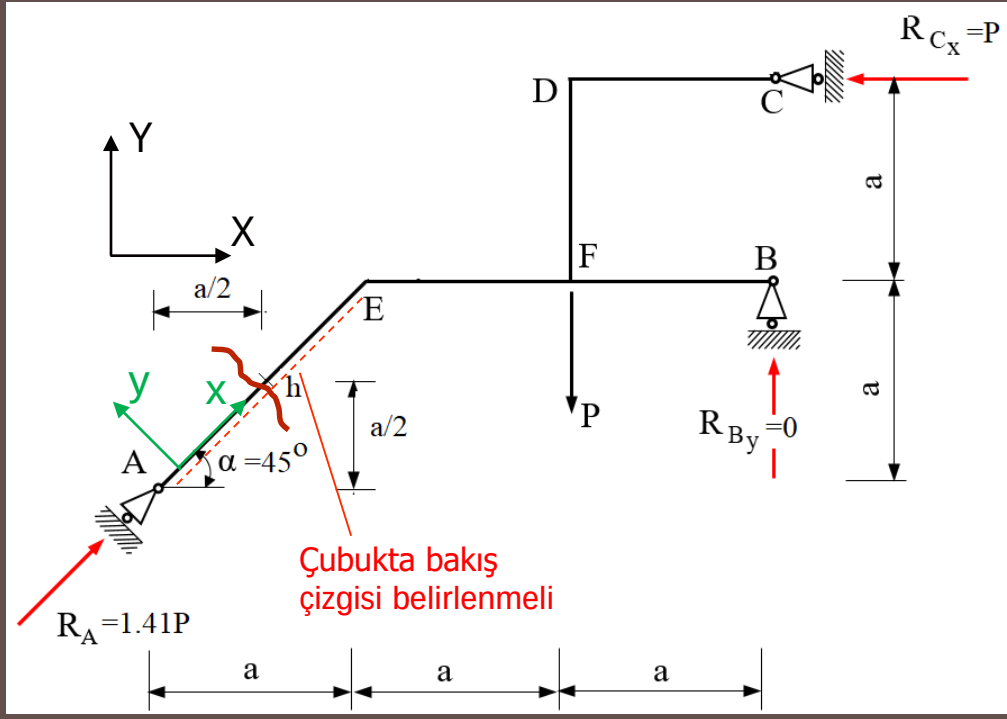


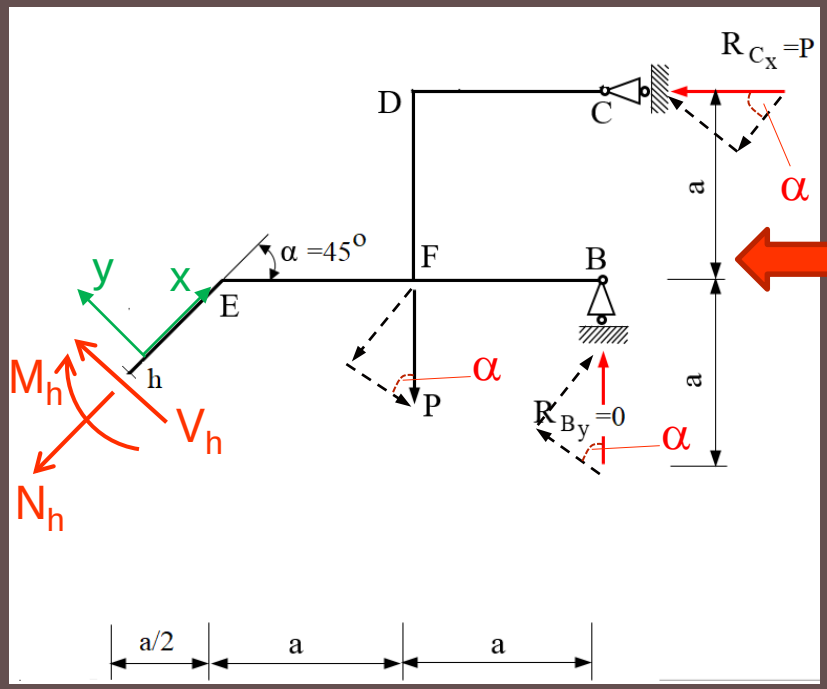
$$\sum x = 0;\ N_h = -1.41P$$

$$\sum y = 0;\ V_h = 0$$

$$\sum M_h = 0;\ M_h = 0$$

II. Seçenek: Kesimin sağında kalan sistem parçası üzerinde eleman eksenlerine göre denge şartları yazılacaktır;





Sistem eksenlerine (X,Y) göre tanımlı tüm kuvvetlerin (tepkiler-yükler) eğik çubuğun eleman eksenlerine (x,y) göre bileşenleri belirlenmelidir.

$$\sum x = 0;\ N_h = -P\sin\alpha - P\cos\alpha = -1.41P$$

$$\sum y = 0;\ V_h = P\cos\alpha - P\sin\alpha = 0$$

$$\sum M_h = 0;\ M_h = -P(a + a/2) + P(a + a/2) = 0$$